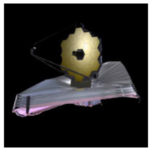

7944 - A MIRI/MRS Deep Dive into Dust Loss and Disk Chemistry for Three Archetypical Proplyds

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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Dr. Thomas Haworth (CoI) (ESA Member)	Queen Mary University of London

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI Observations				
	1	177-341 MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) 177-341
	2	142-301 MIRI/MRS	MIRI Medium Resolution Spectroscopy	(2) 142-301
	3	072-135 MIRI/MRS	MIRI Medium Resolution Spectroscopy	(3) 072-135

ABSTRACT

Most stars and planets form in clusters in the presence of massive stars. This UV-rich environment can drive a substantial loss of gas from protoplanetary disks via a photoevaporative outflow. While Hubble, ALMA, and JWST images have glimpsed the photoevaporating “proplyds” in Orion, there is still much we do not understand about the effect of strong external UV fields on planet formation. How much dust, and what grain sizes, is entrained in the outflowing gas? How does a disk’s warm molecular chemistry respond to a strong external UV field? To answer these questions, we propose deep MIRI/MRS observations of three archetypical proplyds in Orion, enabling a spatial/spectral analysis of the central disk and outflowing material. We will search for mid-IR silicate emission features in the proplyd outflows to measure dust abundance and grain sizes entrained in the photoevaporating gas. The spectra of the central disks will reveal their warm molecular gas inventory, which we will compare with systems in low-mass star-forming regions. We will also search for telltale signs of UV-driven chemistry, e.g., OH and CH3+. Approved cycle 3 NIRSpec IFU measurements of these three systems will yield the temperature and density of the warm outflowing gas, setting the stage to maximize the scientific return from these MIRI observations. Two of the three systems also have existing complementary high-resolution multi-band ALMA data. We propose for ALMA observations of the third systems to reveal its large dust population, cold molecular gas, and the free-free emission from the surrounding ionization front.

OBSERVING DESCRIPTION

The goal is to acquire deep MIRI/MRS observations of three proplyds in the Orion Nebula Cluster. This will enable a search for spatially extended silicate emission features from the proplyd outflow. It will also reveal molecular emission lines from the central disk. We request all three grating modes. We use a 4-point dither pattern optimized for extended sources. We do not request target acquisition. The observations are centered on the proplyd head and central disk; measuring the full extent of the tail is not critical. We optimize the exposures to obtain SNR>250 from 5-18 um on the central disks while avoiding saturation.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	177-341	RA: 05 35 17.7000 (83.8237500d) Dec: -05 23 41.10 (-5.39475d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: The coordinates are set to ensure the proplyd head and central disk are within the FOV of the smallest IFU fields. The end of the proplyd tail will fall outside the FOV. *Category=Star* *Description=[Proplyds, Protoplanetary disks]* *Extended=YES*				
(2)	142-301	RA: 05 35 14.1500 (83.8089583d) Dec: -05 23 0.90 (-5.38358d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: The coordinates are set to ensure the proplyd head and central disk are within the FOV of the smallest IFU fields. The end of the proplyd tail will fall outside the FOV. *Category=Star* *Description=[Proplyds, Protoplanetary disks]* *Extended=YES*				
(3)	072-135	RA: 05 35 7.2200 (83.7800833d) Dec: -05 21 34.30 (-5.35953d) Equinox: J2000	Epoch of Position: 2000	
Comments: The coordinates are set to ensure the proplyd head and central disk are within the FOV of the smallest IFU fields. The end of the proplyd tail will fall outside the FOV. *Category=Star* *Description=[Proplyds, Protoplanetary disks]* *Extended=YES*				

Observation

Proposal 7944, Observation 1: 177-341 MIRI/MRS

Fri Aug 08 23:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Data Excess over lower threshold

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	177-341	RA: 05 35 17.7000 (83.8237500d) Dec: -05 23 41.10 (-5.39475d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: The coordinates are set to ensure the proplyd head and central disk are within the FOV of the smallest IFU fields. The end of the proplyd tail will fall outside the FOV.
Category=Star
Description=[Proplyds, Protoplanetary disks]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	BRIGHTSKY	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	9	100	1	Dither 1	4	400	3457.659	217232.25
1	SHORT(A)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.1
1	SHORT(A)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.1
2		IMAGER	F770W	FASTR1	5	150	1	Dither 1	4	600	3111.547	217232.25
2	MEDIUM(B)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.2
2	MEDIUM(B)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.2
3		IMAGER	F1000W	FASTR1	5	150	1	Dither 1	4	600	3111.547	217232.25
3	LONG(C)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.9
3	LONG(C)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.9

Observation

Proposal 7944, Observation 2: 142-301 MIRI/MRS

Fri Aug 08 23:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	142-301	RA: 05 35 14.1500 (83.8089583d) Dec: -05 23 0.90 (-5.38358d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: The coordinates are set to ensure the proplyd head and central disk are within the FOV of the smallest IFU fields. The end of the proplyd tail will fall outside the FOV.
Category=Star
Description=[Proplyds, Protoplanetary disks]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	BRIGHTSKY	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	9	100	1	Dither 1	4	400	3457.659	217232.25
1	SHORT(A)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.1
1	SHORT(A)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.1
2		IMAGER	F770W	FASTR1	5	150	1	Dither 1	4	600	3111.547	217232.25
2	MEDIUM(B)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.2
2	MEDIUM(B)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.2
3		IMAGER	F1000W	FASTR1	5	150	1	Dither 1	4	600	3111.547	217232.25
3	LONG(C)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.9
3	LONG(C)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.9

Observation

Proposal 7944, Observation 3: 072-135 MIRI/MRS

Fri Aug 08 23:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Data Excess over lower threshold

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	072-135	RA: 05 35 7.2200 (83.7800833d) Dec: -05 21 34.30 (-5.35953d) Equinox: J2000	Epoch of Position: 2000	

Comments: The coordinates are set to ensure the proplyd head and central disk are within the FOV of the smallest IFU fields. The end of the proplyd tail will fall outside the FOV.
Category=Star
Description=[Proplyds, Protoplanetary disks]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	BRIGHTSKY	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	9	100	1	Dither 1	4	400	3457.659	217232.25
1	SHORT(A)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.1
1	SHORT(A)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.1
2		IMAGER	F770W	FASTR1	5	150	1	Dither 1	4	600	3111.547	217232.25
2	MEDIUM(B)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.2
2	MEDIUM(B)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.2
3		IMAGER	F1000W	FASTR1	5	150	1	Dither 1	4	600	3111.547	217232.25
3	LONG(C)	MRSLONG		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.9
3	LONG(C)	MRSSHORT		FASTR1	15	20	1	Dither 1	4	80	3540.951	222303.9